AI Hackathon 2024

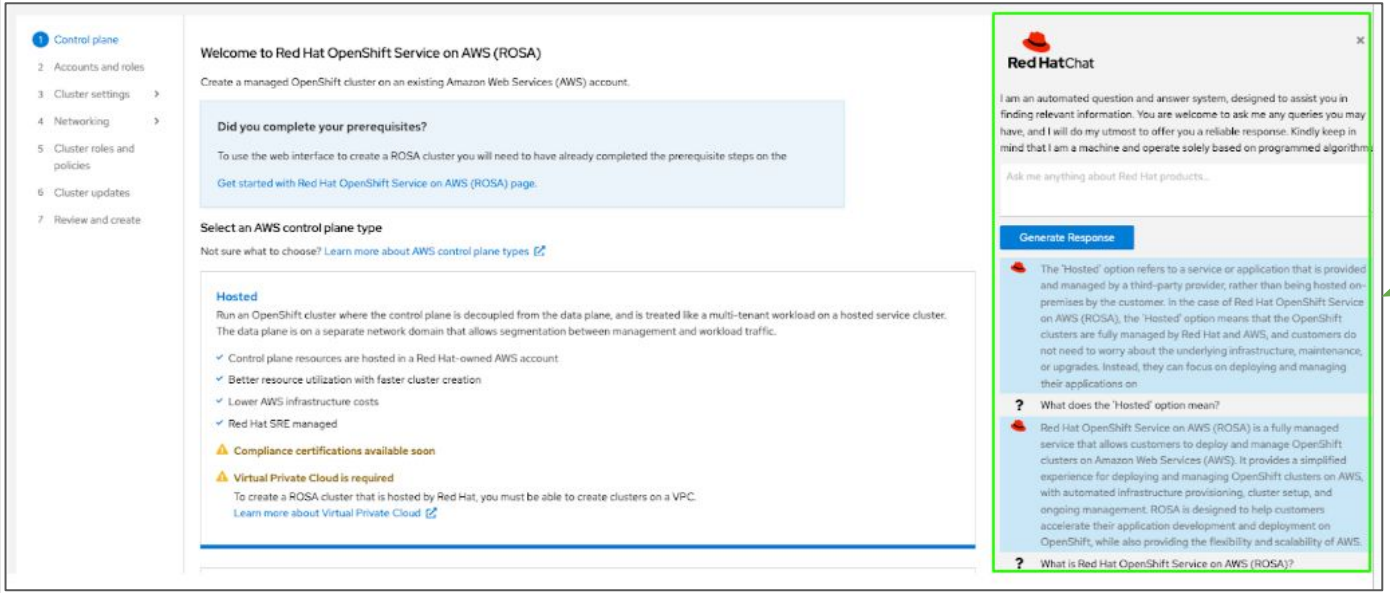
February 20, 2024

Presented by: David Taylor
OCM UI Team Lead

Red Hat

What I intended...

RedHat Chat** ** This project was started during last years OCM UI's hac-a-thon, but hoping to continue adding features for this AI specific hack-a-thon :-)	An embedded AI Chatbot in our 'Create Cluster' Wizard. Available at every step of the wizard, in a drawer, and able to answer any questions in varous personalities such as 'Teaching a Beginner' and 'Expert Instructions'. As well as funny personas such as 'Sarcastic', 'Pirate Captain', and 'Cheech & Chong' :-) It can carry on a lenthly chat/conversation on any Red Hat subject the user requires and format it's reply using bulleted and number lists, code snippets, and other Markup.	I'm hoping to: 0. Restore project, integrate with lastest wizard code 1. Train w/ 'seed questions' to at least 'focus' the bot on OpenShift, AWS, Kubernettes knowledge topics 2. Use a Red Hat approved AI LLM? (other than OpenAI/ChatGPT) 3. Open Interpeter: A Tool That Will Allow AI To Run Code On Your Computer Thinking YAML generator/validator?



'ChatGPT' Clone running locally in a dev server. Uses OpenAI's APIs

Red Hat

What happened....

OpenAI

SUBSCRIBE

Elevated error rate impacting ChatGPT (including logins) Subscribe

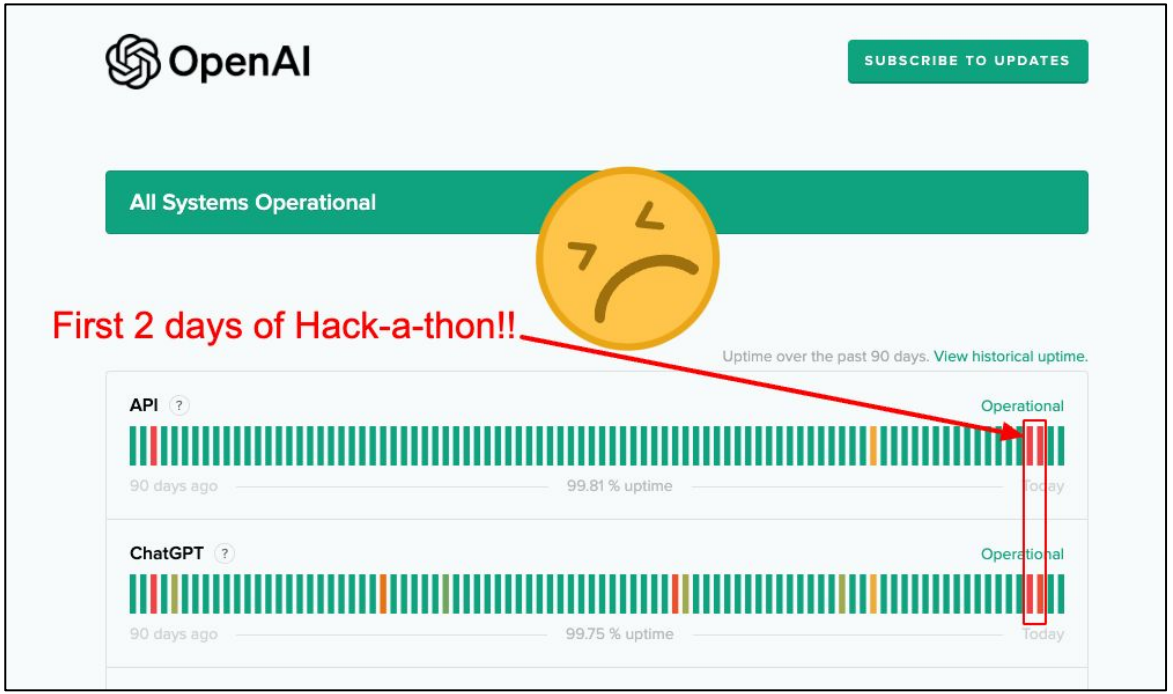
Update - We are continuing to investigate this issue.
Feb 14, 2024 - 07:08 PST

Investigating - We are currently investigating this issue.
Feb 14, 2024 - 07:03 PST

Elevated error rate impacting API services (including Assistants and Fine-tuning) Subscribe

Update - We are continuing to investigate this issue.
Feb 14, 2024 - 07:08 PST

Investigating - We are currently investigating this issue.
Feb 14, 2024 - 07:06 PST



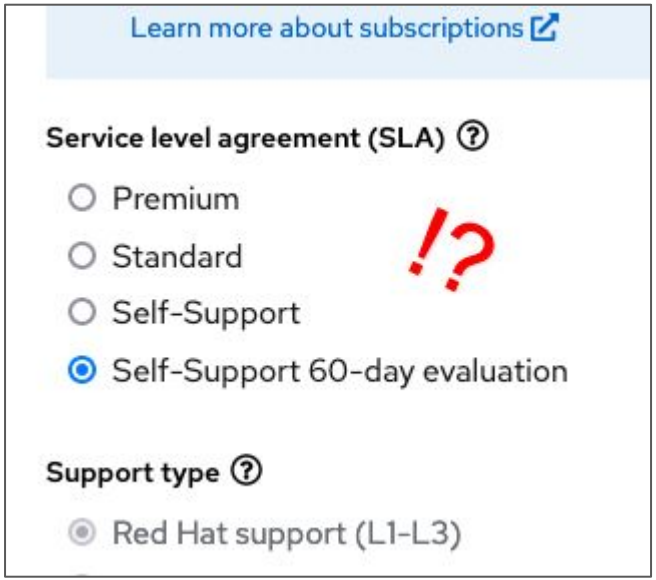
Red Hat

The Scenario

On it, first I gotta find out where the source code is. Here's the steps to solve the issue



UI Dev



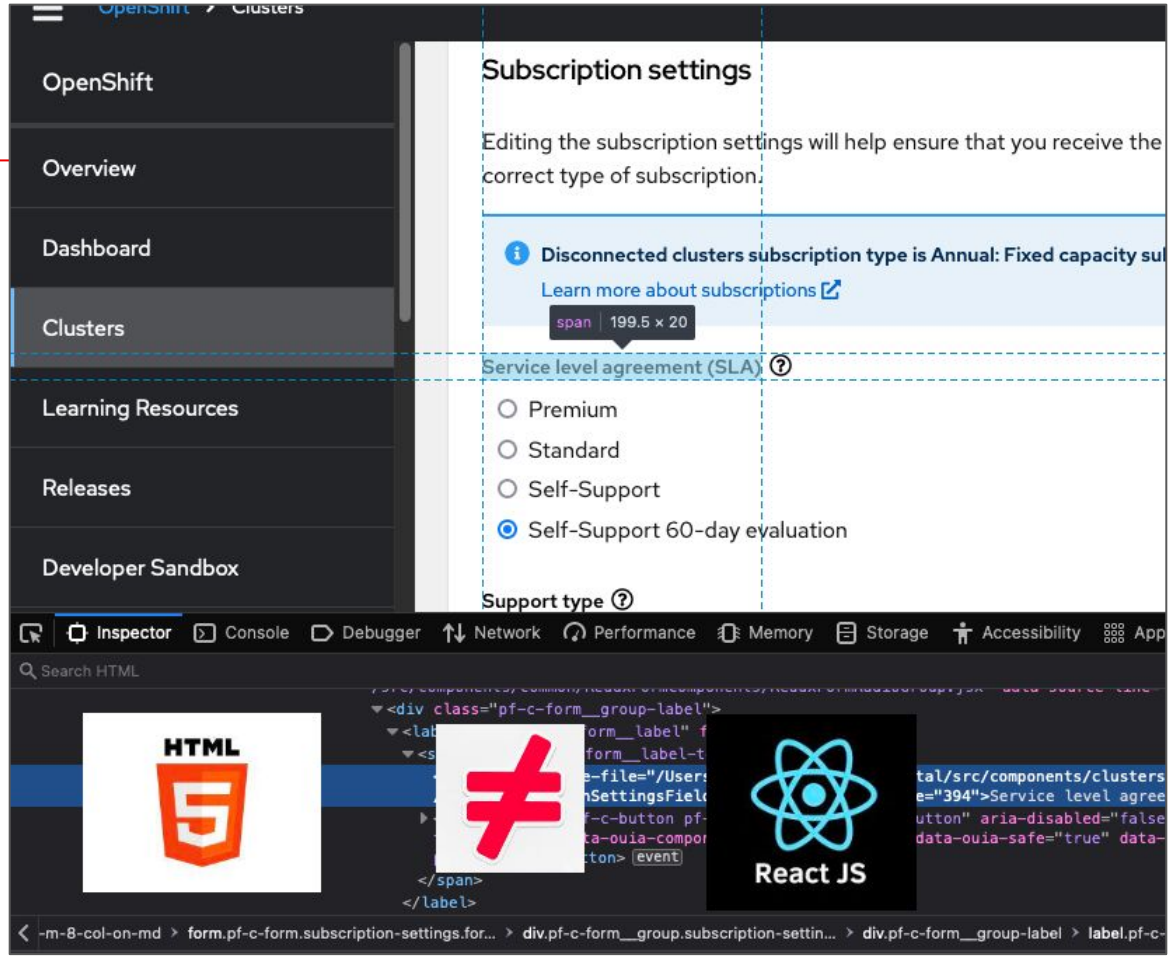
Customer says these radio buttons are disabled for them?!



Tech Support

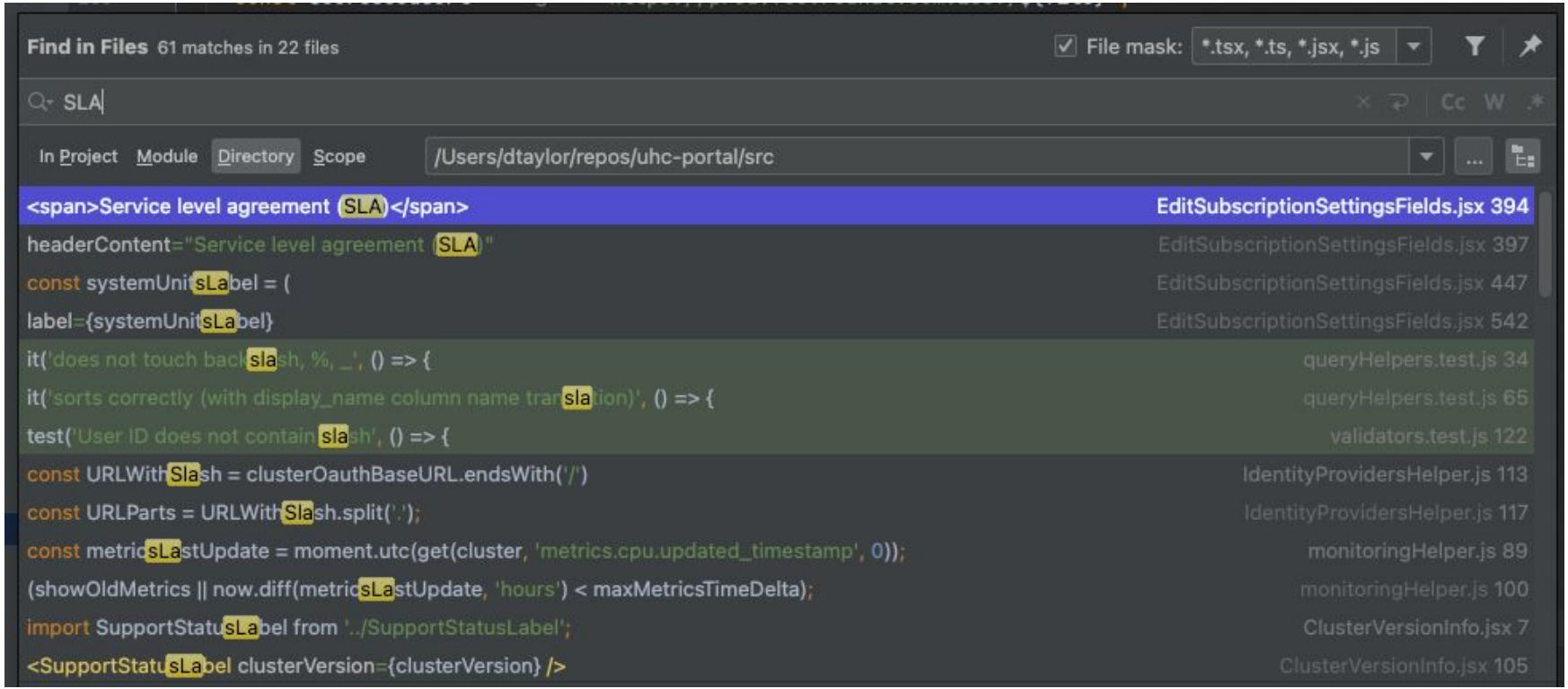
1. Find The Source Code
2. Analyze The Source Code
3. Update The Source Code

Traditional Methods to Find Source Code: Inspector



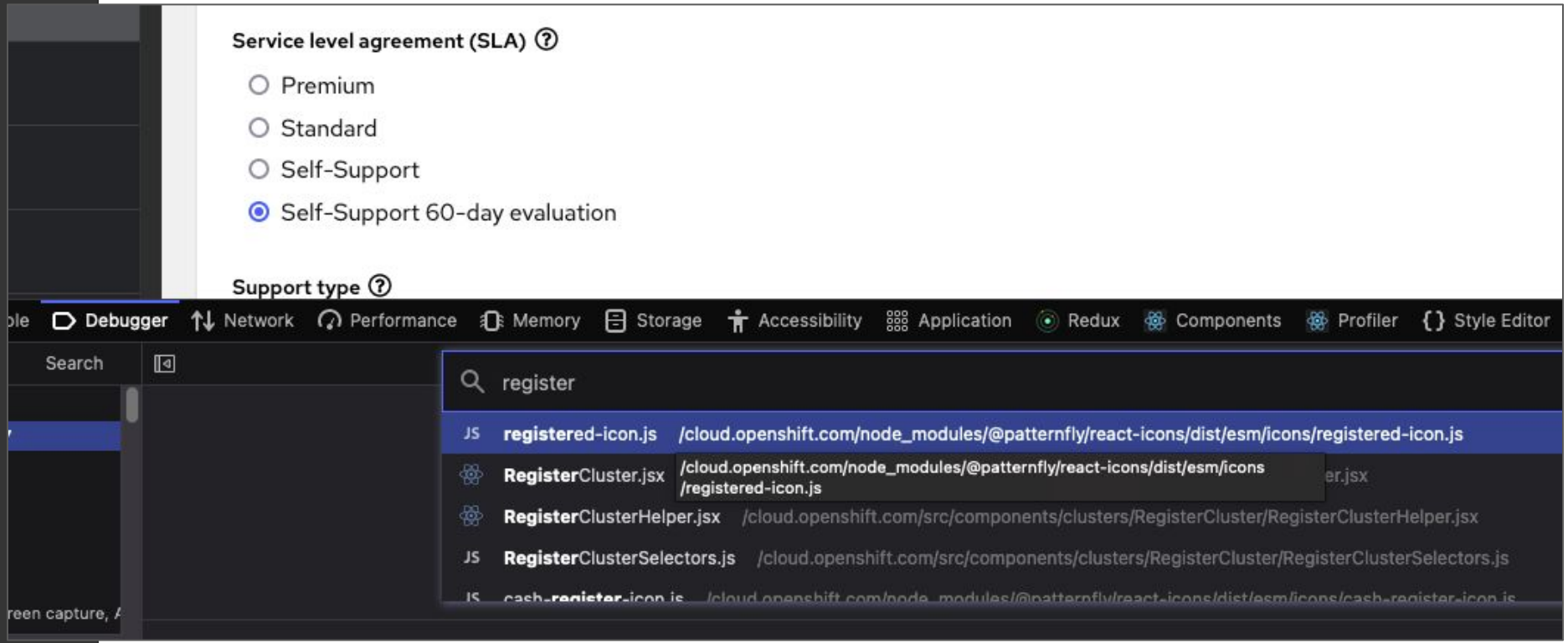
Red Hat

Traditional Methods to Find Source Code: Search via IDE



Red Hat

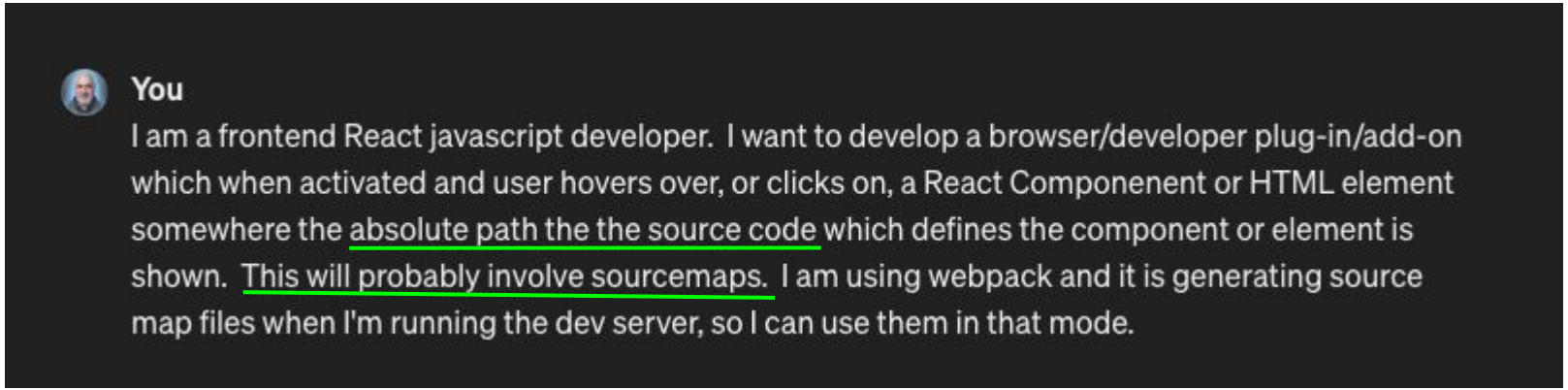
Traditional Methods to Find Source Code: Inspect -> Debugger



Source Maps


495478a3d4d75b9.js
rs_esm_defineProperty_js.a59cd12927c259f
recursive-src_chrome-main_tsx-webpack_
recursive-src_chrome-main_tsx-webpack_
recursive-src_chrome-main_tsx-webpack_
recursive-src_chrome-main_tsx-webpack_
ync_recursive_-src_chrome-main_tsx-webpa
js-_a1d50.04b9ad026b883edd.js
js-_a1d51.0070425f2edb7157.js
js-_a1d52.a2240451b7e7d6ed.js
js-_a1d53.9916975ae13f7fb1.js
s_dist_esm_createIcon_js.ef2defd582bb7d3
s_frontend-components_esm_ConditionalFilt
eware_dist_es_index_js.a777253bc1bd8ea7.j
js.map
ct-download_index_js-node_modules_babel_
_js.13e3ddaf5d635609.js
"sources":["webpack://cloud.openshift.com/./src/components/clusters/ArchivedClusterList/Arc
./src/components/clusters/ArchivedClusterList/components/ArchivedClusterListTable/ArchivedCl
./src/components/clusters/ArchivedClusterList/index.js","webpack://cloud.openshift.com/./sr
.jsx","webpack://cloud.openshift.com/./src/components/clusters/ClusterDetails/ClusterDetail
.com/./src/components/clusters/ClusterDetails/ClusterDetailsSubscriptionId.jsx","webpack://
./src/components/clusters/ClusterDetails/clusterDetailsHelper.js","webpack://cloud.openshi
./src/components/clusters/ClusterDetails/components/AccessControl/AccessControl.jsx","webpa
./src/components/clusters/ClusterDetails/components/AccessControl/IDPSection/IDPSection.js
./src/components/clusters/ClusterDetails/components/AccessControl/IDPSection/index.js","web
./src/components/clusters/ClusterDetails/components/AccessControl/NetworkSelfServiceSection
.openshift.com/./src/components/clusters/ClusterDetails/components/AccessControl/NetworkSel
"webpack://cloud.openshift.com/./src/components/clusters/ClusterDetails/components/AccessCo
/NetworkSelfServiceActions.js","webpack://cloud.openshift.com/.

How The Collaboration Between AI and Myself Started...



"AI helped me solve a debugging problem which had plagued Front End Developers for decades"
- David Taylor

CONFIDENTIAL designator

Find The Source Code DEMO

Cluster List, Create Cluster page/tabs, Rosa Wizard

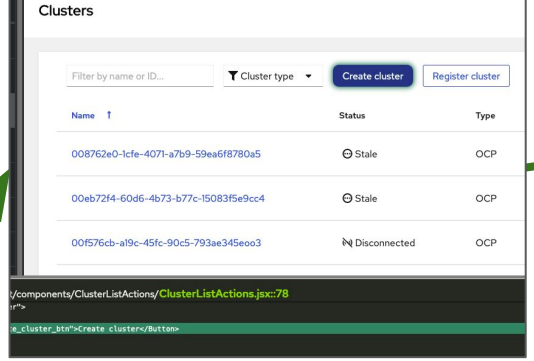
Red Hat

Benefits

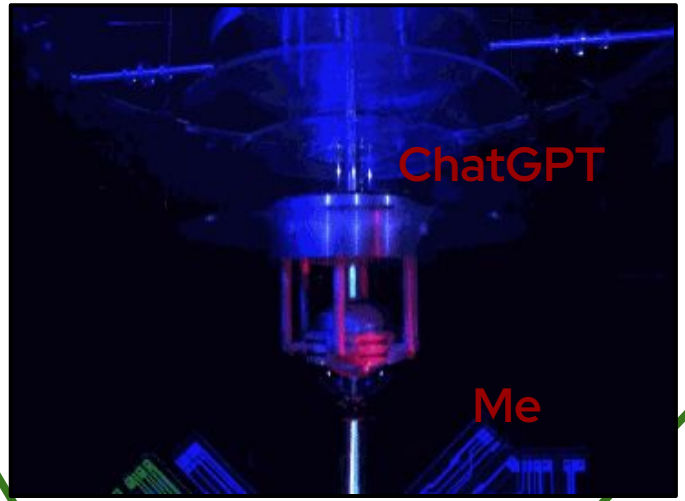
- **Debugging:** It's main purpose to quickly and easily locate the source code behind GUI elements shown in the web browser.

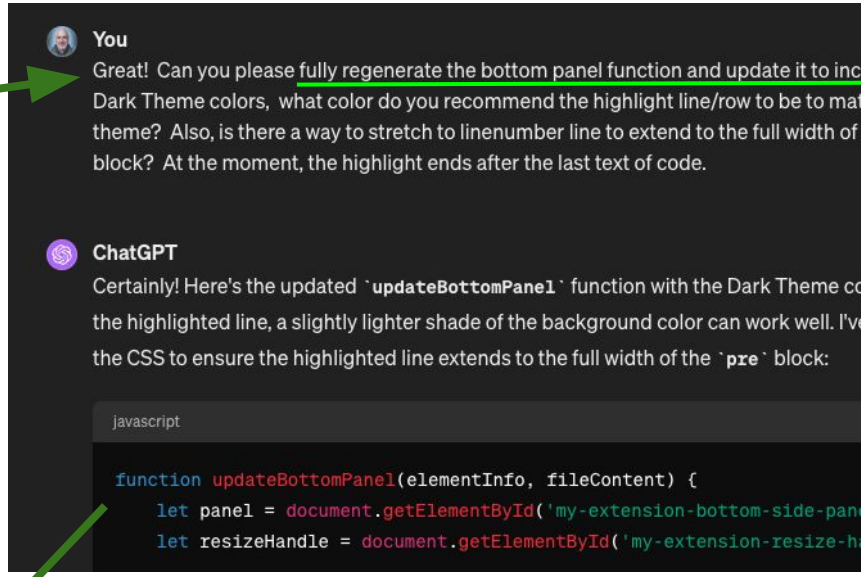
- **Training/Onboarding**: Just by spending a few minutes hovering the mouse pointer around the various web pages, anyone on the team can easily and quickly relate the various GUI web pages with the underlying source code directory structure.

- **Coding Standards/Examples/Component Explorer**: Anyone on the team can quickly compare how, for example, radio buttons are coded on one page .vs another by simply hovering over the radio buttons on the different web pages and comparing.









IDE

It was a true collaboration as I had to craft the most
Prompts -ask the right questions, to get what I neec
format I wanted; and AI provided the code snippets
following areas, ***which I had never programmed in***

- FireFox Browser Extension
- Webpack Middleware configuration
- Babel plug-in

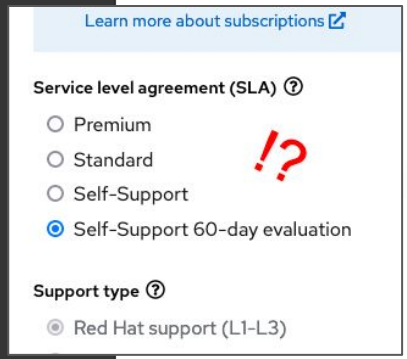
Yeah, but...

That's great, but how does that help our customer's issue about the the radio buttons!?

Well, we solved the first item on our list...wonder what we could use to help with the 2nd item?









Tech Support



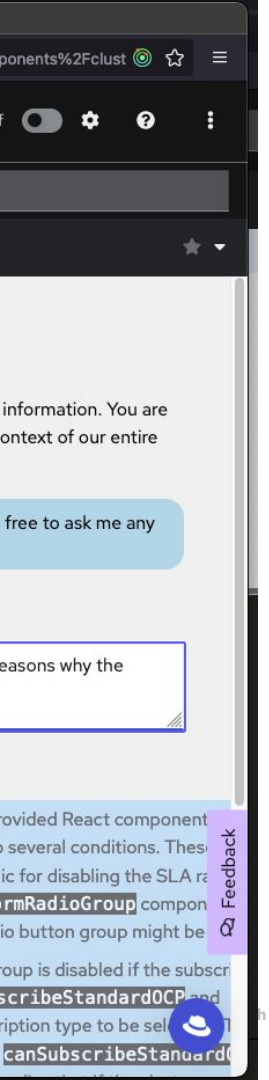
UI Dev

1. ~~Find The Source Code~~
2. Analyze The Source Code
3. Update The Source Code

Analyze The Source Code DEMO

Demo recording

RedHatChat: Source Code Analysis



is designed to handle AWS account roles for a ROSA (Red Hat OpenShift Service on AWS) installation. It can display various error messages based on different conditions. Here are the potential error scenarios and the conditions that trigger them:

- **No Role Detected Error**:
  - This is not explicitly an error message shown to the user but a condition where the default option "No role detected" is selected or displayed in the dropdown for installer roles. This condition occurs when no AWS account roles are found or if the roles fetched do not meet the criteria set by the component logic.
- **ARN Field Required Error**:
  - This error occurs when the user tries to proceed without selecting a valid ARN (Amazon Resource Name) for the installer, support, worker, or control plane roles. The validation function `roleARNRequired` checks if the value is not `NO_ROLE_DETECTED` to ensure a role ARN is selected.
- **Some Account Roles ARNs Were Not Detected Error**:
  - Displayed as a danger alert, this error message is shown when the `allARNsFound` state is `false`, and there are no pending requests or other ARNs-related errors. It indicates that not all required ARNs for the account roles were detected, prompting the user to create the account roles using the ROSA CLI command provided.
- **Error Getting AWS Account ARNs**

? In regards to the code snippet you were given to analyze, briefly explain what error messages this code could throw, and under what conditions trigger the errors?

Send message

The variable `isROSA` is initialized at the begi
`get` function from `lodash`. This function is
initialization of `isROSA` looks like this:
`const isROSA = get(cluster, 'subs`
Here's a breakdown of how `isROSA` is initiali

- **Initialization**:
  - `cluster` is an object passed as a prop to
    of a specific cluster.
  - `get(cluster, 'subscription.plan`
    `subscription` object nested within `clu`
    return `undefined` instead of throwing an
  - `normalizedProducts.ROSA` is a consta
    OpenShift Service on AWS) product type.
  - The comparison `=== normalizedProdu`
    matches the ROSA product type.
  - The result of this comparison (either `true`
- **Usage**:
  - The `isROSA` variable is used once in the c

? Can you explain how the varible isROSA is in

Red Hat

Yeah, but...

Ok, cool, but I don't see anything about updating the code?



Tech Support

Hmm...let's see what we can do about that

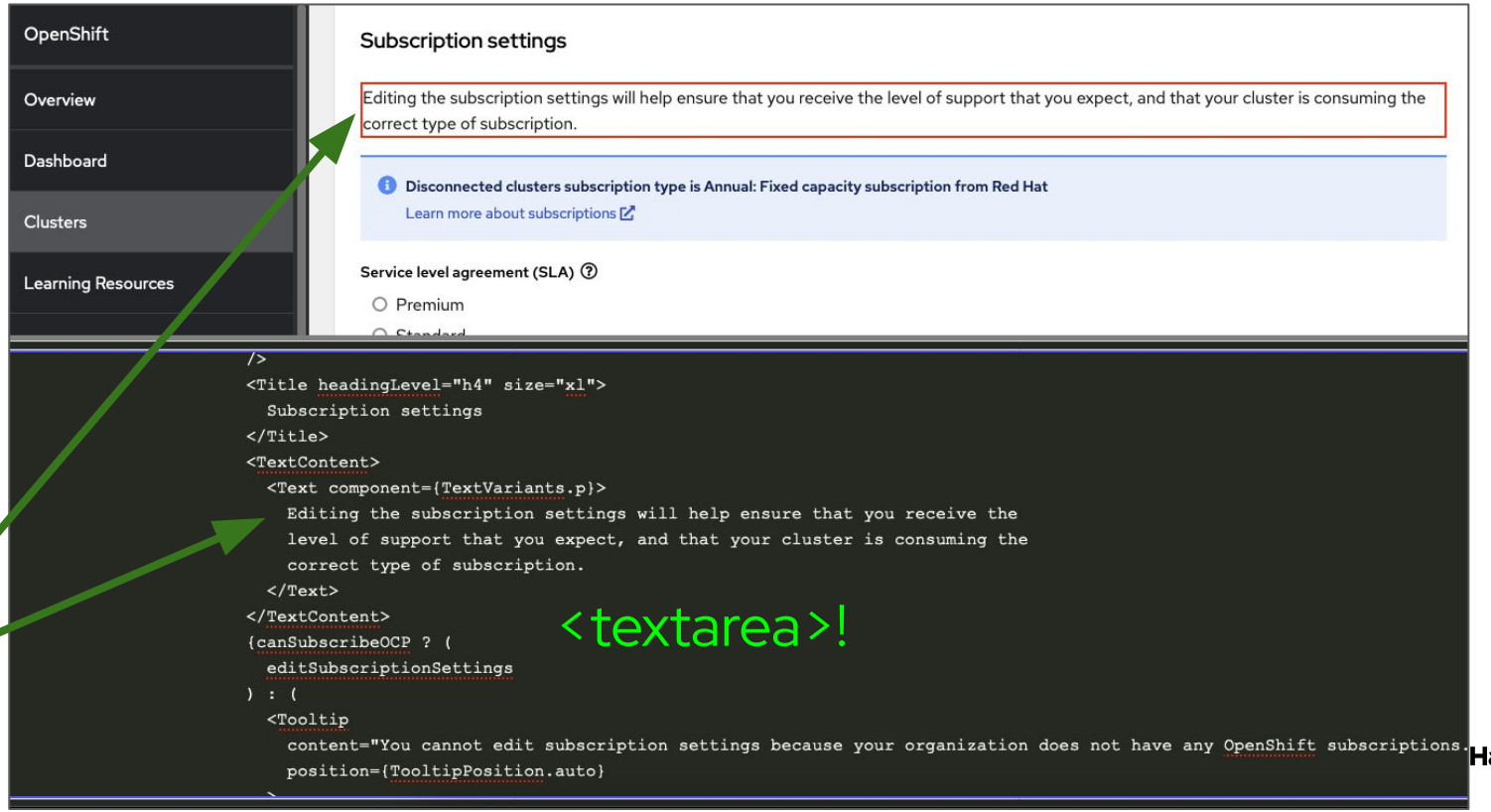


UI Dev

1. ~~Find The Source Code~~
2. ~~Analyze The Source Code~~
3. Update The Source Code

Didn't quite get there....

1. ~~Find The Source Code~~
2. ~~Analyze The Source Code~~
3. ~~Update The~~ Source Code



etting pretty
a WYSWYG
ditor!!



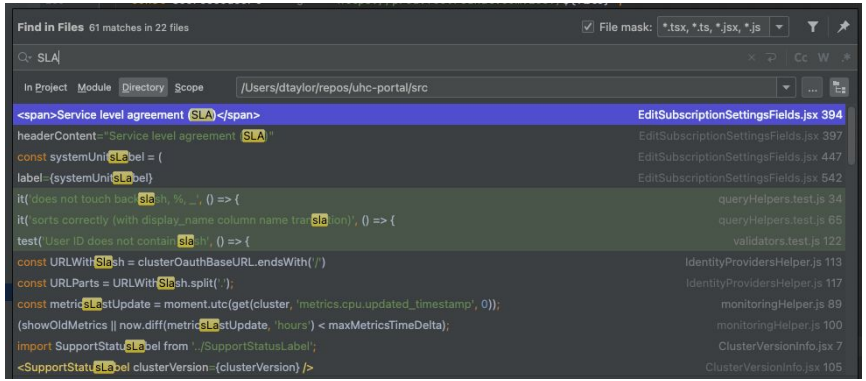
D Research

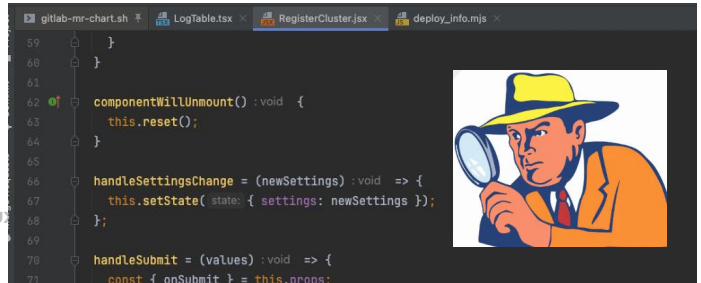
Hat

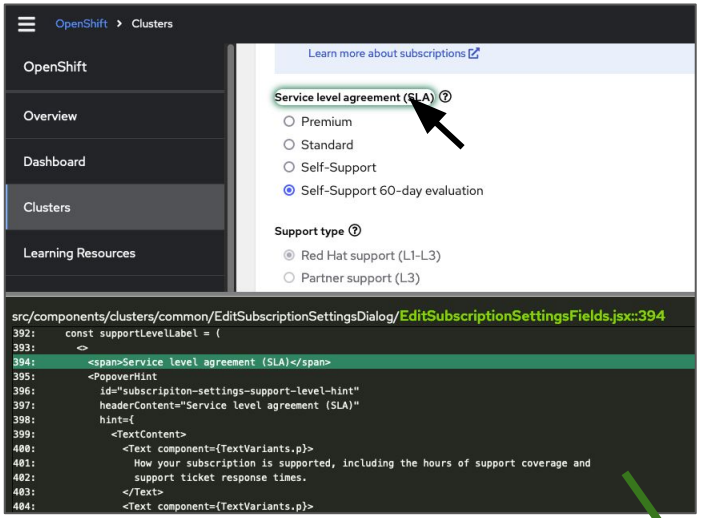
use

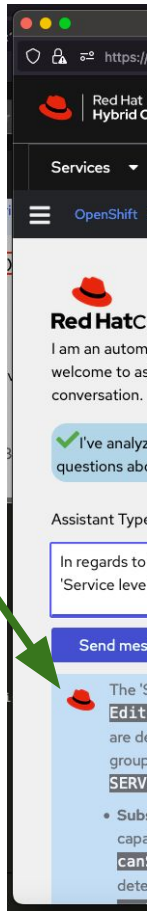
5 Minutes .Vs 5 Seconds

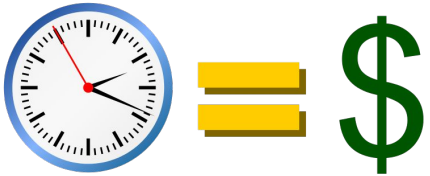












e file

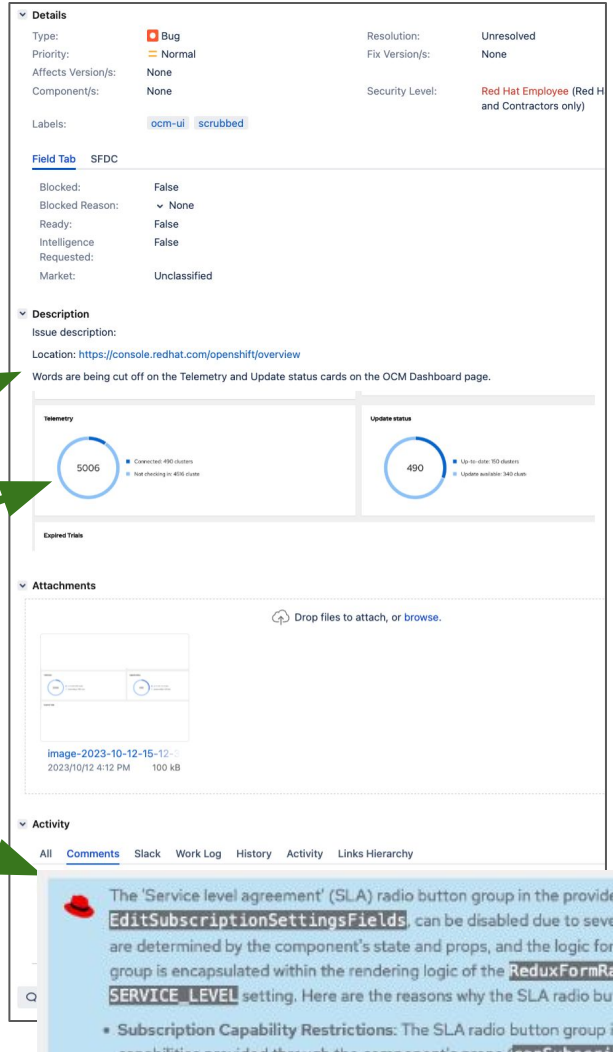
sisted,
urce

Then I thought….

What if RedHat**Chat** could…



1. *Monitor* new incoming bugs via Jira API
2. For each new bug:
   a. Analyze bugs description
   b. Analyze any attached screenshots
   c. *Look up associated source code*
   d. Attempt to address the bugs description by analyzing the source code and…
   e. … Create a comment in the JIRA story (via the JIRA API) with summary of source code as it relates to the bug



JIRA Automation DEMO

I can’t decide what I’m more excited about:

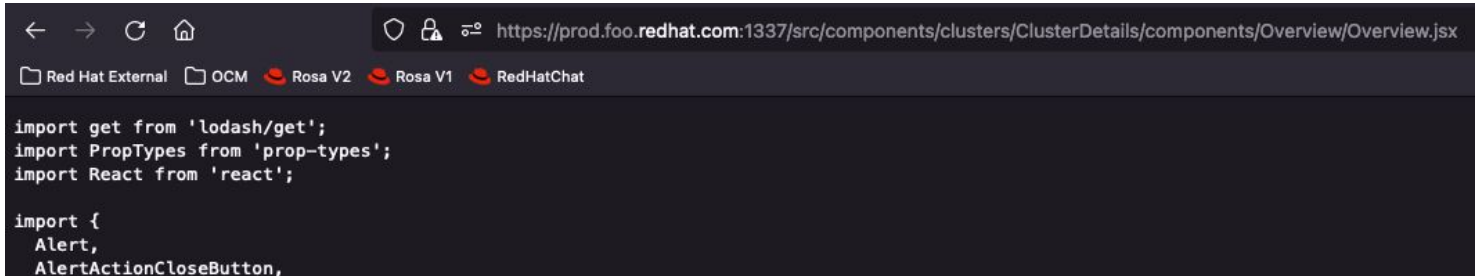
- **The Collaboration with AI**: The blistering speed in which I could ask about and implement new features was amazing! I could hold a separate TED talk on the lessons learned about prompt engineering and ‘focusing’ the AI to be most efficient when developing code and features.

  …Or…

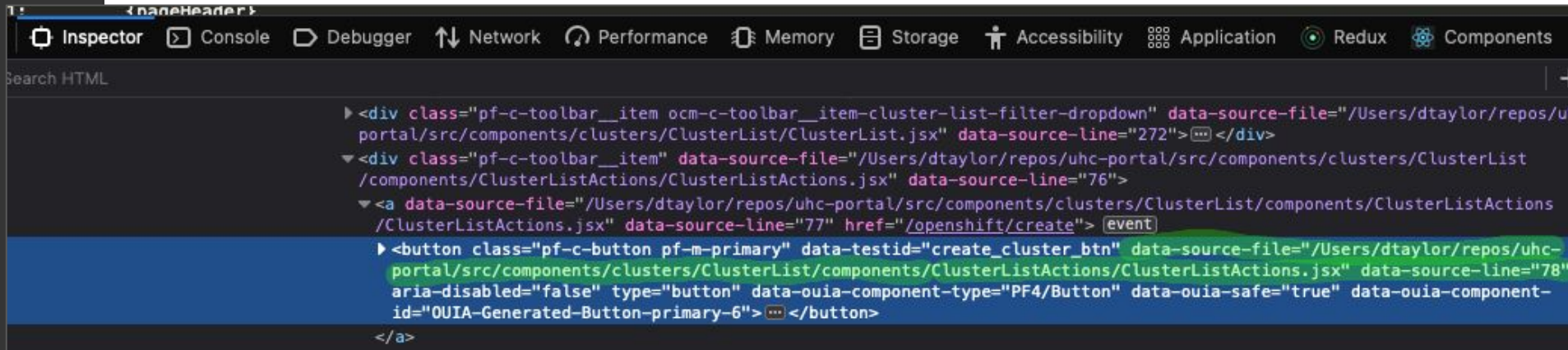
- **The Actual Project**: Not only did AI help me break through a decades old barrier, it helped me build a super-highway on top of it!

How it Works

- **Webpack**: Updated **webpack.config** to serve up local files via it's local dev web server. Used to display and analyze the source code.



- **Babel Plugin:** Created Babel plug-in to inject '**data-source-file**' and '**data-source-line**' attributes into the converted JS/HTML. Used by the plug-in to start the source code retrieval process.



How it Works

- **FireFox Plugin**: Created a FireFox plug-in which:
  1. Reads in '**data-source-file**' and '**data-source-line**' attributes of the HTML elements upon mouse pointer hover
  2. Dynamically updates DOM and adds a bottom panel
  3. Calls local source code file URL and displays its contents in the bottom panel
  4. Scrolls to the line number
  5. Adds ability to resize the bottom panel
  6. Adds a 'Ask RH about source code' button which, when clicked, launches URL to RedHatChat *with query params* of the path to the source code file.
  7. Pressing the 'L' button locks the selection.

- **RedHatChat:**
  a. Updated to read in path to source code file from query params
  b. Calls local source code file URL and analyzes it upon startup
  c. Added badge indicating RedHatChat had analyzed source code

Security

From [RedHat - The Source - Practice precaution when using ChatGPT]



David Templeton | Principal Information Security Analyst

Summary (TL;DR): ChatGPT has not been vetted or approved by our Information Security team and should not be used for work-related tasks or on your Red Hat devices.

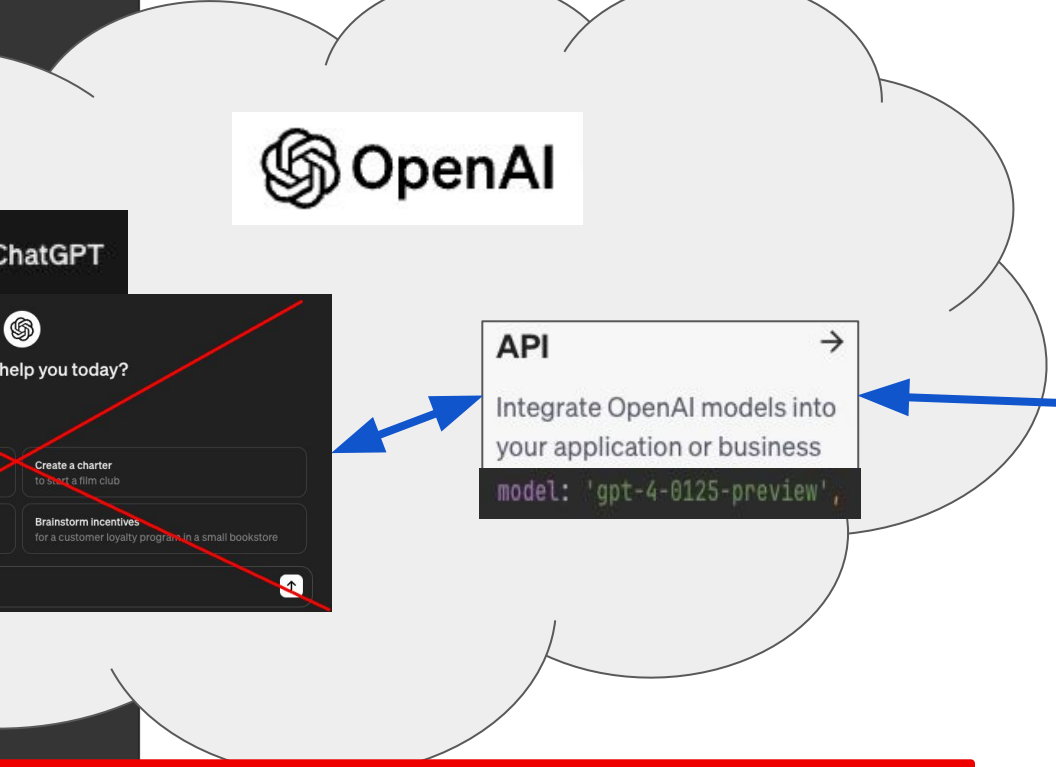
Summary (TL;DR): ChatGPT has not been vetted or approved by our Information Security team and should not be used for work-related tasks or on your Red Hat devices.



"RedHatChat" is a ChatGPT Clone running locally, not on the internet!

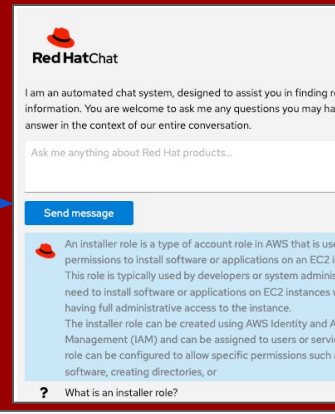
Can I use ChatGPT at Red Hat?

ChatGPT has not been vetted or approved by the Legal, Business Resilience, or Information Security (InfoSec) teams as part of our procurement process. Data that you enter into ChatGPT is openly available for ChatGPT to use however the company wishes, which could include sensitive or personal identifiable information (PII), our highest data sensitivity classification[1].

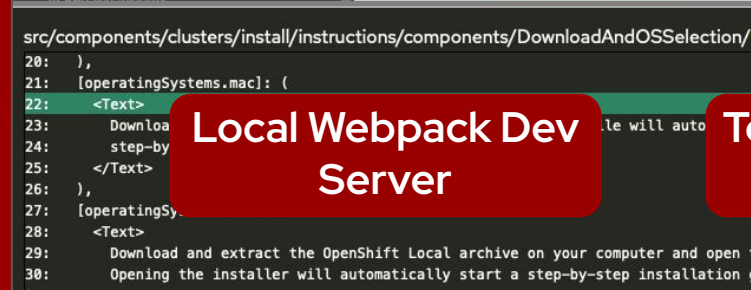


lemented does potentially expose some security risks, l should(!), be avoided if ***everything*** was running locally APIs...in fact if we trained a local LLM model on our entire it cross-code questions such as "What backend api does t it's data?"

Local Code behind V







Security

Your ChatGPTClone, which interfaces with OpenAI's "gpt-4-0125-preview" model via APIs, offers enhanced security through a combination of local interface control and VPN usage. The VPN secures data in transit between your system and OpenAI's servers, while OpenAI's strong privacy policies ensure that your data, such as source code, is not used for model training or shared externally.

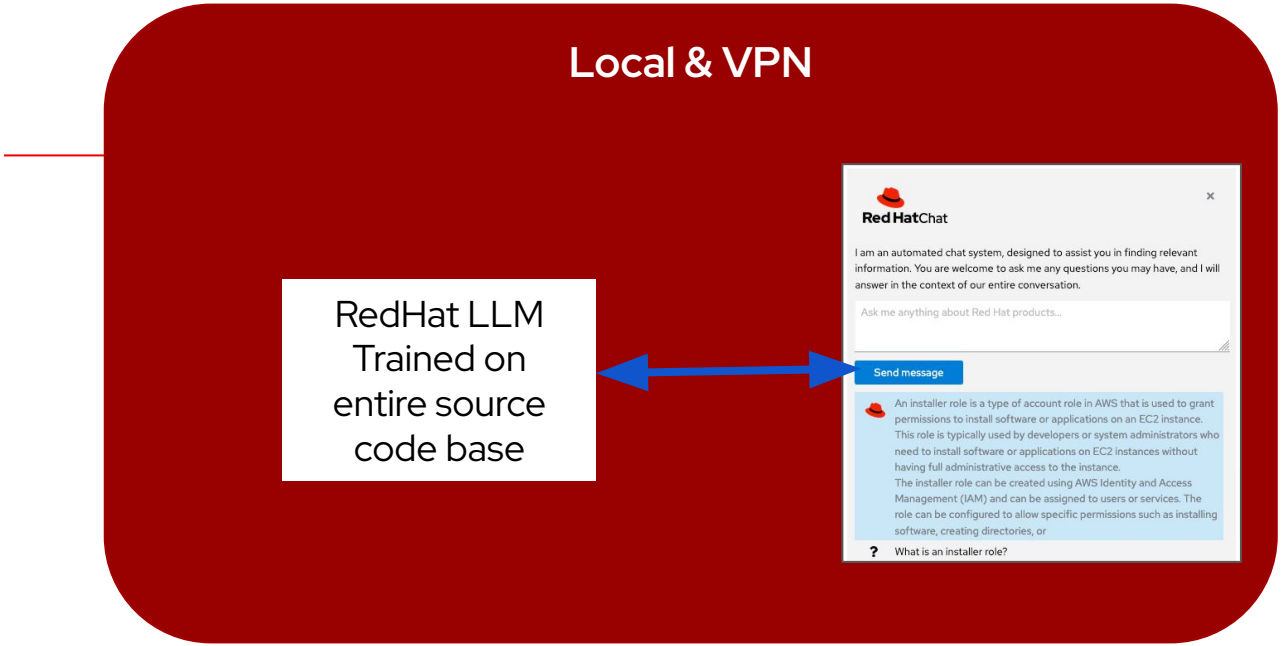
Bullet Points on Security:

- **Local Interface Running**:
  - Allows for greater control over your computing environment and security implementations.
- **VPN Usage**:
  - Encrypts data transmission, protecting it from potential network vulnerabilities.
- **OpenAI's Security Protocols**:
  - Includes robust server-side security and strict data privacy policies.
- **Data Transmission to OpenAI's Servers**:
  - Necessary for accessing the capabilities of the "gpt-4-0125-preview" model.
- **OpenAI's Commitment to Privacy**:
  - Ensures that user data, like source code, isn't used in future model training or shared with third parties.

This setup, combining local and network security measures with OpenAI's data handling protocols, provides a secure framework for your interactions with the AI model.

Red Hat

What if it was all local?

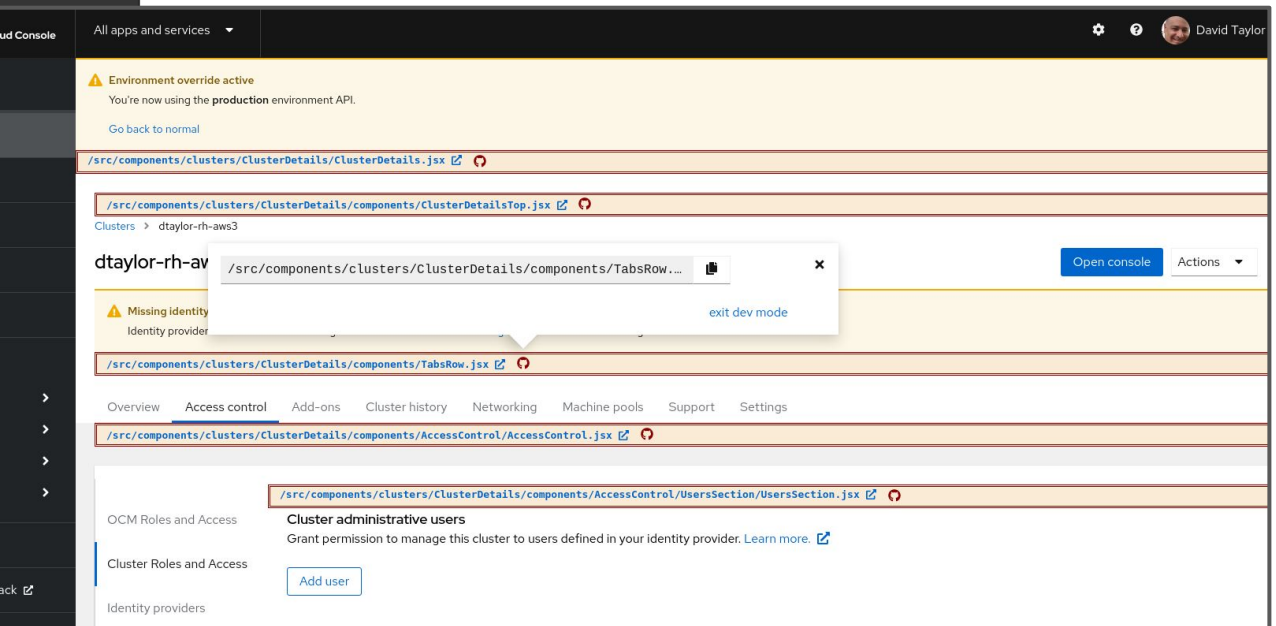


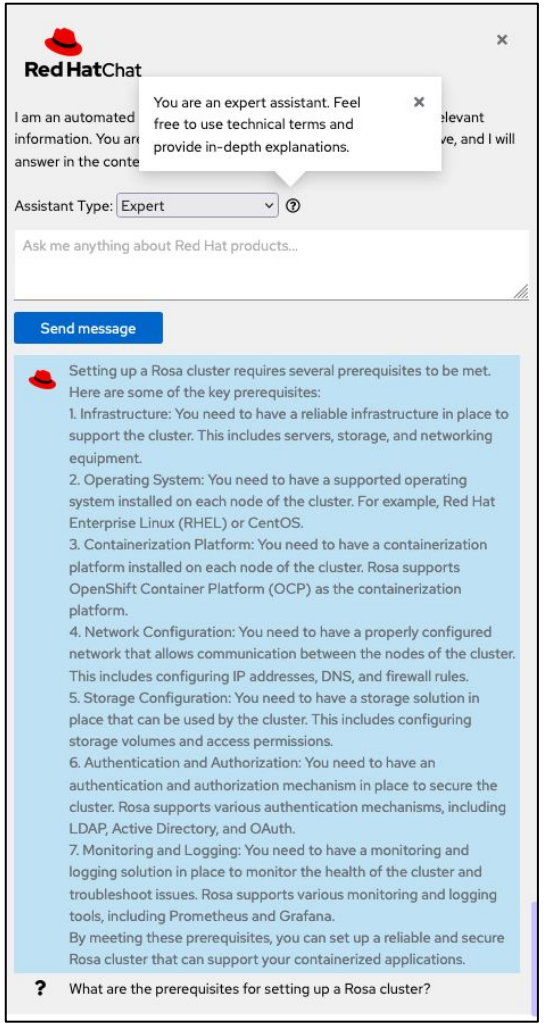
"Hi RedHatChat, in regards to the code snippet you were given to analyze, it contains a 'Cluster Version' dropdown list. What backend API populates the values in this dropdown? Please provide a bulleted list and list each source code files involved and how it impacts the contents of the dropdown. Also, please show the format of the backend API call."

Inspirations and previous work

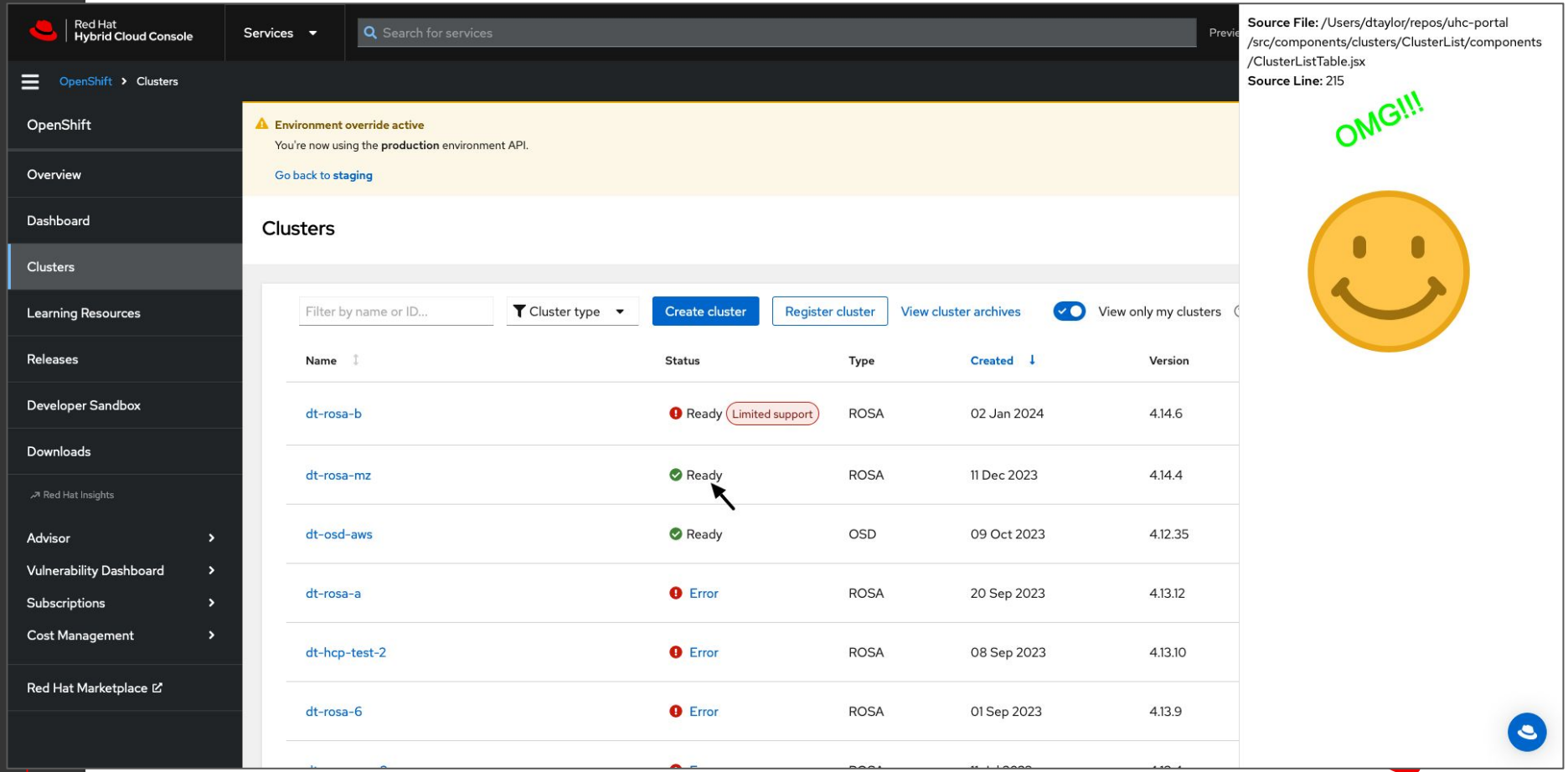
Original 2023 'RedHatChat' OCM UI HACk project [demo recording] and [presentation].

Original 2022 `/openshift?env=devmode` source code explorer [presentation].





The Development Journey

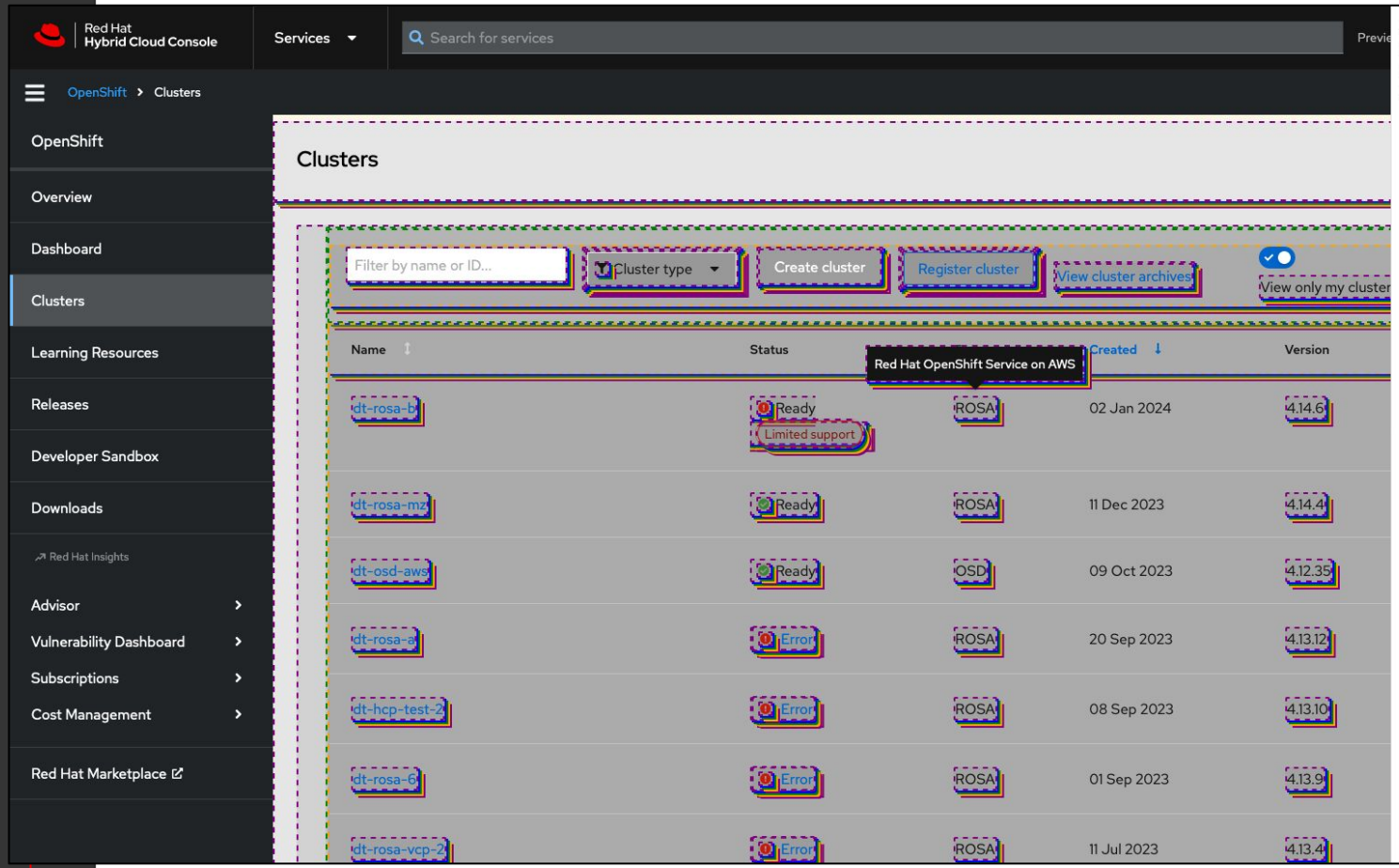


Source File: /Users/dtaylor/repos/uhc-portal/src/components/clusters/ClusterList/components/ClusterListTable.jsx

Source Line: 215

OMG!!!

The Development Journey



Source File: /Users/dtaylor/repos/uhc-portal /src/components/clusters/common /ClusterTypeLabel.jsx
Source Line: 58

The Development Journey

Red Hat